Albian **SALIHU**

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Swiss national · French (native) · Albanian (native) · English (C1)

Education

EPFL, École Polytechnique Fédérale de Lausanne, Lausanne Sept **2017** – Sept **2024**
MSc Robotics (minor Space Technologies) · BSc Microengineering · Preparatory year (CMS)

CPLN, Centre Professionnel du Littoral Neuchâtelois, Neuchâtel Sept **2012** – Aug **2017**
Federal Vocational Baccalaureate · Federal VET Diploma (CFC) in Electronics

Professional experience and projects

Wasteflow — Robotics & Systems Integration Engineer May **2025** – Present

- Piloting an industrial PoC at a paper-sorting facility, owning systems integration and commissioning; developed real-time control adjusting a 4-conveyor chain based on upstream contamination data from the vision infrastructure, achieving a 20% increase in throughput.
- Trained and deployed waste-classification models for live decision-making; iterated using controlled lab runs and field data.

Swiss MotionTech — Master Thesis Feb **2024** – Sept **2024**

- Designed a real-time closed-loop feedback system correcting for multi-hardness silicone 3D printing. Validated on prosthetic liners in production.

EPFL Spacecraft Team — Semester Project Sept **2023** – Jan **2024**

- Designed and implemented a dual-mode filament controller in VHDL on a Xilinx Zynq UltraScale+, regulating the thermionic emitter of a CubeSat TOF mass spectrometer for the ESA-supported CHESS mission, in collaboration with the University of Bern.

EPFL Robot Competition 2023 — Team member Feb **2023** – Jun **2023**

- Owned electronics integration for an autonomous ROS 2 robot: motor controllers, sensor wiring (Lidar, IMU, camera), electrical schematics, Raspberry Pi 4 and Coral TPU integration.

Technical and computer skills

- **Core strengths:** Real-time control systems, embedded and FPGA development, computer vision and AI integration, including coursework in Spacecraft design and system engineering and Space mission design and operations.
- **Embedded & FPGA:** VHDL, Xilinx Zynq UltraScale+, Intel Quartus/ModelSim, AXI4-Lite, hardware/software co-design, PID control, real-time signal acquisition and actuation.
- **Robotics & Control:** ROS 2, UAV control theory, sensor integration, PLC integration.
- **AI & Data:** Machine learning and deep learning pipelines, model training and deployment for industrial vision tasks.
- **Electronics & CAD:** PCB design and prototyping, circuit assembly (soldering, crimping), CAD tools (Fusion 360, Catia, Inventor, FreeCAD).
- **Programming:** Python (PyTorch, scikit-learn), C/C++ (OpenCV), VHDL, MATLAB, C#, Java.

Interests

Astronomy — I follow space missions and planetary science closely; I enjoy stargazing on clear nights.
Trekking — Regular hikes with my dog, good for the head after a week in front of screens.
DIY — Whether it's electronics, furniture, or whatever breaks first, I prefer to understand and fix things myself.
Volunteering — Co-founded Graines d'Avenir, a nonprofit offering academic support and mentoring to children and teens in Neuchâtel.